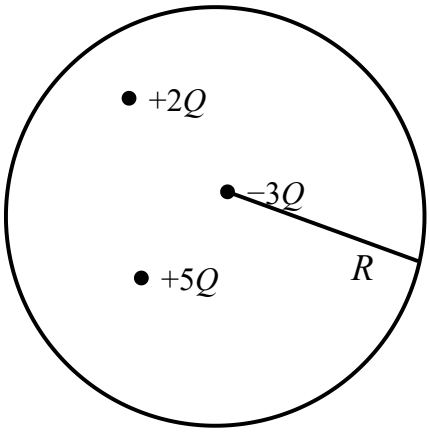
STPM 2015 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. A spherical surface of radius R containing charges $+2Q$, $-3Q$ and $+5Q$ is shown in the diagram below.



What is the total electric flux which passes through the spherical surface?

A $\frac{Q}{\pi\varepsilon_0 R^2}$ B $\frac{Q}{\pi\varepsilon_0 R}$ C $\frac{Q}{\varepsilon_0}$ D $\frac{4Q}{\varepsilon_0}$

Answer: D

Net charge enclosed $= 2Q - 3Q + 5Q = 4Q$; $\Phi = \frac{4Q}{\varepsilon_0}$

Explanation: The flux does not depend on the radius or on where the charges are inside.

2. The work done to bring a positive charge of 1.6×10^{-19} C from infinity to a point P in an electric field is 8.0×10^{-16} J. What is the electric potential at P?

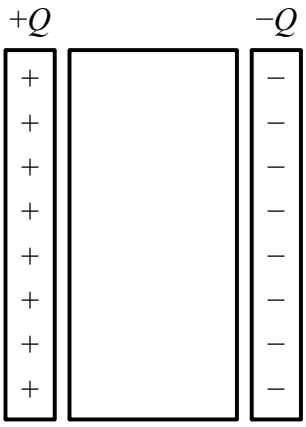
A -5.0×10^3 V B -2.5×10^3 V C 2.5×10^3 V D 5.0×10^3 V

Answer: D

$$V = \frac{W}{q} = \frac{8.0 \times 10^{-16}}{1.6 \times 10^{-19}} = 5.0 \times 10^3 \text{ V}$$

Explanation: Positive work was needed, so the potential is positive.

3. A parallel plate capacitor with dielectric is shown in the diagram below.



The charges on the capacitor plates are $+Q$ and $-Q$. Which statement is **not** true about the capacitor?

A The electric field within the dielectric is uniform.
B The electric field between the plates is dependent on the net surface charge density.
C The induced charges on the surface of the dielectric are equal in magnitude to the charges of the capacitor plates.
D The electric field due to the induced charges is in the opposite direction to the electric field due to the charges on the capacitor plates.

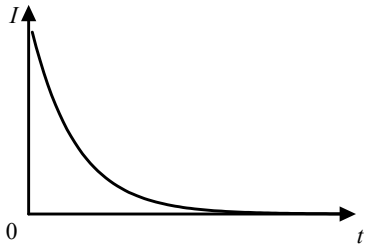
Answer: C

The induced surface charges on the dielectric are smaller than the charges on the plates (otherwise the field inside would be zero), so statement C is not true.

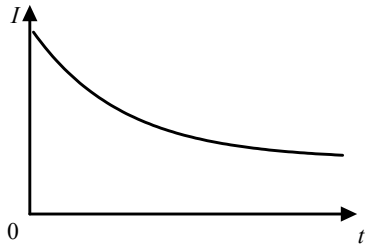
Explanation: The induced charge is $Q(1 - 1/\varepsilon_r)$.

4. A capacitor is connected in series with a switch, a resistor of low resistance and a battery. Which graph represents the variation of the current I with time t during the charging process?

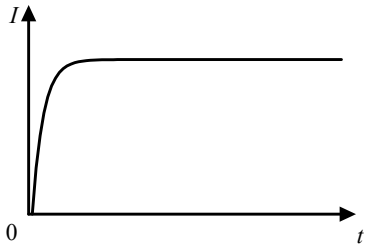
A



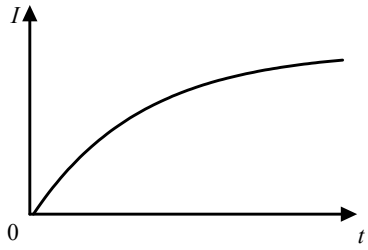
B



C



D



Answer: A

The charging current starts at its maximum and decays exponentially; with a low resistance (small RC) it falls to zero very quickly (A).

Explanation: $I = I_0 e^{-t/RC}$.

5. Drift velocity of free electrons in a metal conductor is

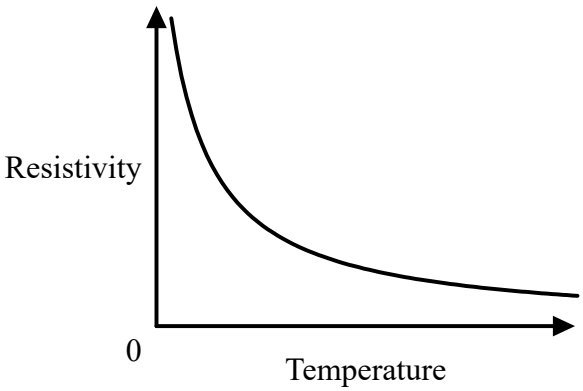
A the most probable speed of the free electrons in the conductor
B the root mean square speed of the free electrons in the conductor
C the mean velocity of the free electrons in the conductor which is due to electric field
D the mean velocity of the free electrons in the conductor which is due to its random motion

Answer: C

Drift velocity is the mean velocity of the free electrons produced by the electric field (C).

Explanation: The random motion alone averages to zero.

6. The variation of the resistivity of a material with temperature is shown in the graph below.



The most likely material is

A non-pure metal B semiconductor C superconductor D good conductor

Answer: B

The resistivity falls steeply as the temperature rises: a semiconductor (B).

Explanation: Metals show a resistivity that increases with temperature.

7. Which is **not** true about the shunt used to convert a galvanometer into an ammeter?

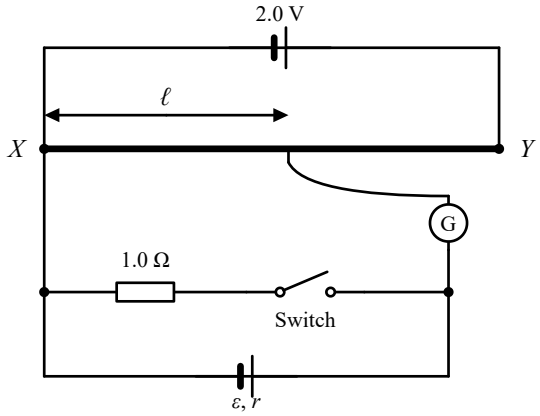
A It has a very low resistance.
B It prevents current from flowing through the galvanometer.
C It is always connected in parallel with the galvanometer.
D It diverts part of the current from flowing through the galvanometer.

Answer: B

A shunt only diverts most of the current; some current must still pass through the galvanometer to deflect it. Statement B is not true.

Explanation: The shunt has a low resistance and is in parallel with the galvanometer.

8. A potentiometer with a slide wire XY of length 100.0 cm is connected to a battery of e.m.f. 2.0 V and is used to determine the e.m.f. ε and the internal resistance r of another battery as shown in the diagram below.



When the switch is opened, the balanced length ℓ is 60.0 cm and when the switch is closed, the balanced length ℓ is 32.0 cm. What are the values of ε and r?

	ε	r
A	0.64 V	0.88 Ω
B	0.64 V	1.88 Ω
C	1.20 V	0.88 Ω
D	1.20 V	1.88 Ω

Answer: C

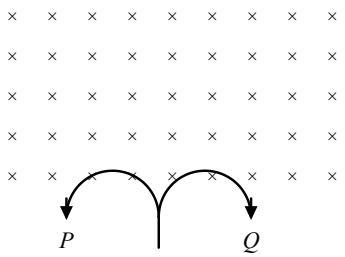
$$\text{Gradient} = \frac{2.0}{100.0} = 0.020 \text{ V cm}^{-1}\text{; switch open: } \varepsilon = (0.020)(60.0) = 1.20 \text{ V}$$

$$\text{Switch closed: } V = (0.020)(32.0) = 0.64 \text{ V}, I = \frac{0.64}{1.0} = 0.64 \text{ A}, r = \frac{1.20 - 0.64}{0.64} = 0.88 \text{ Ω}$$

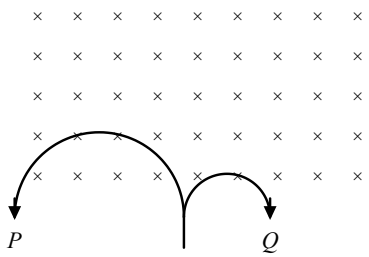
Explanation: With the switch open no current flows from the cell at balance.

9. A positively charged particle P and a negatively charged particle Q are projected into a magnetic field at the same speed. If the magnitude of the charges are the same and the masses $m_P = 2m_Q$, which diagram shows the correct paths of P and Q in the magnetic field?

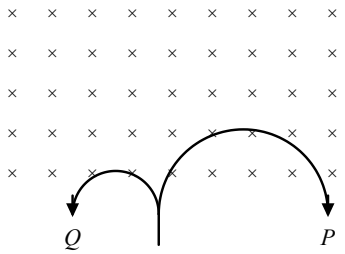
A



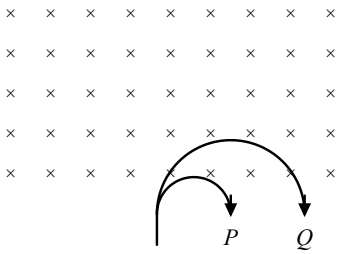
B



C



D



Answer: B

With **B** into the page and both particles moving up, the positive P curves to the left and the negative Q to the right.

$$r = \frac{mv}{qB} \propto m\text{: } P \text{ has twice the radius of } Q \text{ (B)}$$

Explanation: Same speed and same size of charge.

10. Which statement is true about the magnetic force acting on a straight current-carrying conductor which is placed in a uniform magnetic field?

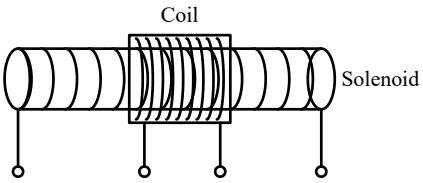
A The magnetic force is independent of the length of the conductor.
B The magnetic force will cause the conductor to move in a circular motion.
C The magnetic force is maximum when the conductor is placed parallel to the magnetic field.
D The magnetic force always acts in the perpendicular direction to the magnetic field and the conductor.

Answer: D

$\mathbf{F} = I\mathbf{L} \times \mathbf{B}$ is always perpendicular to both the field and the conductor (D).

Explanation: It is maximum when the conductor is perpendicular to the field, and zero when parallel.

11. A coil is wound over the middle part of a long solenoid as shown in the diagram below.



The mutual inductance between the solenoid and the coil is 2.43×10^{-3} H. If the current in the solenoid is increased at the rate of 14 A s^{-1}, what is the magnitude of the e.m.f. induced in the coil?

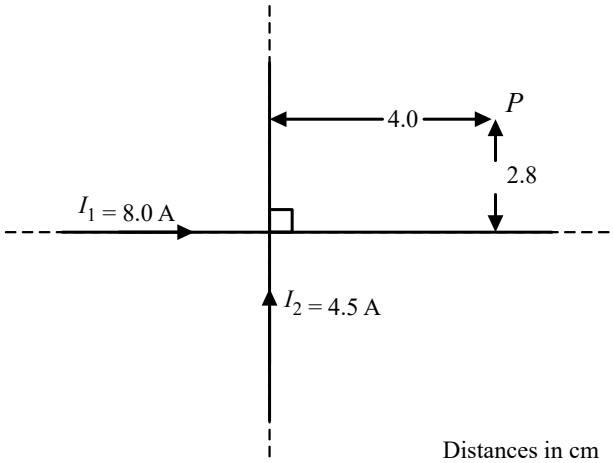
A 1.7×10^{-4} V B 1.7×10^{-2} V C 3.4×10^{-2} V D 5.8×10^{3} V

Answer: C

$$\varepsilon = M\frac{\mathrm{d}I}{\mathrm{d}t} = (2.43 \times 10^{-3})(14) = 3.4 \times 10^{-2} \text{ V}$$

Explanation: Mutual induction.

12. Two thin current-carrying conductors of infinite length are fixed perpendicularly to each other as shown in the diagram below.



The resultant magnetic flux density at point P is

A 3.5×10^{-5} T B 8.0×10^{-5} T C 1.1×10^{-4} T D 2.5×10^{-4} T

Answer: A

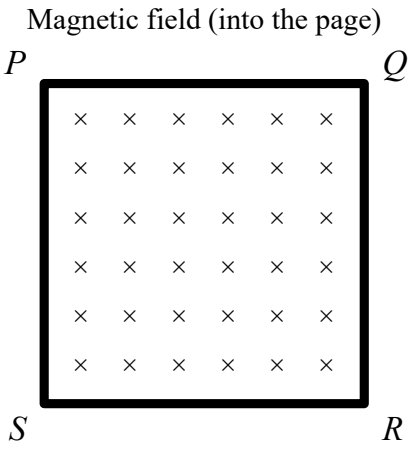
From I_1 (P is 2.8 cm above it): $\dfrac{\mu_0(8.0)}{2\pi(0.028)} = 5.71 \times 10^{-5}$ T, out of the page

From I_2 (P is 4.0 cm to its right): $\dfrac{\mu_0(4.5)}{2\pi(0.040)} = 2.25 \times 10^{-5}$ T, into the page

Resultant = $5.71 - 2.25 = 3.5 \times 10^{-5}$ T (out of the page).

Explanation: Use the right-hand grip rule for each wire.

13. A metal frame *PQRS* of area 4.0×10^2 cm^2 and resistance 4.0 Ω is placed in a uniform magnetic field of 2.0 T as shown in the diagram below.



If the magnetic field decreases to zero at a constant rate in 0.50 s, what is the magnitude and direction of the induced current?

	Magnitude	*Direction*
A	40 mA	*PQRS*
B	40 mA	*PSRQ*
C	20 mA	*PQRS*
D	20 mA	*PSRQ*

Answer: A

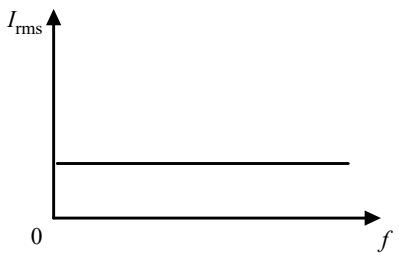
$$\varepsilon = A\frac{\Delta B}{\Delta t} = (4.0 \times 10^{-2})\frac{2.0}{0.50} = 0.16 \text{ V}; I = \frac{0.16}{4.0} = 40 \text{ mA}$$

The flux into the page decreases, so the induced current is clockwise to maintain it: *PQRS*.

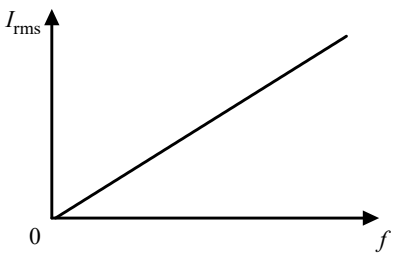
Explanation: 4.0×10^2 cm^2 = 4.0×10^{-2} m^2.

14. A capacitor is connected to an a.c. power supply. If the frequency *f* of the power supply is varied but its r.m.s. voltage is maintained, which graph shows the variation of the r.m.s. current I_{rms} with the frequency *f*?

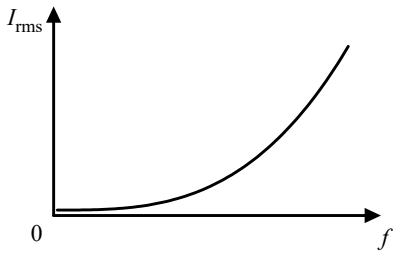
A



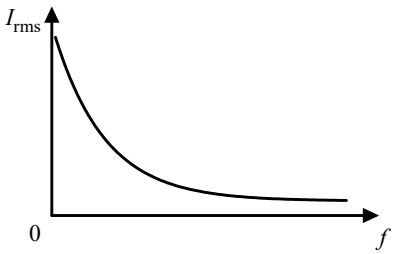
B



C



D



Answer: B

$$I_{rms} = \frac{V_{rms}}{X_C} = V_{rms}(2\pi fC) \propto f\text{: a straight line through the origin (B)}$$

Explanation: A capacitor passes high frequencies more easily.

15. A pure inductor and a pure resistor are connected in series with a sinusoidal voltage supply of maximum voltage 330 V. If the maximum potential difference across the resistor is 200 V, the maximum potential difference across the inductor is

A 92 V B 130 V C 185 V D 262 V

Answer: D

$$V_0^2 = V_R^2 + V_L^2 \Longrightarrow V_L = \sqrt{(330^2 - 200^2)} = 262 \text{ V}$$

Explanation: V_R and V_L are 90° out of phase.

Section B [15 marks]

16. (a) A parallel plate capacitor of surface area 0.24 m^2 has plate separation of 0.65 cm in free space and potential difference of 0.80 kV. Determine the charge stored in the capacitor. [3]

Answer

$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(0.24)}{0.65 \times 10^{-2}} = 3.27 \times 10^{-10}$ F — 1

$Q = CV = (3.27 \times 10^{-10})(800)$ — 1

$Q = 2.61 \times 10^{-7}$ C — 1

3

Explanation: 0.80 kV = 800 V.

(b) The charged capacitor in (a) is then discharged through a circuit connected to a resistor of 10 kΩ. Determine

(i) the time taken for the charge in the capacitor to decrease to half of its initial value, [2]

Answer

$\tau = RC = (10 \times 10^3)(3.27 \times 10^{-10}) = 3.27 \times 10^{-6}$ s — 1

$t = RC \ln 2 = 2.27 \times 10^{-6}$ s — 1

2

Explanation: The half-life of a discharge is $0.693RC$.

(ii) the current after 2.0 μs. [3]

Answer

$I_0 = \frac{V}{R} = \frac{800}{10 \times 10^3} = 0.080$ A — 1

$I = I_0 e^{-t/RC} = 0.080\ e^{-2.0/3.27}$ — 1

$I = 0.043$ A — 1

3

Total for question **8**

Explanation: The current decays with the same time constant as the charge.

17. (a) A battery with e.m.f. ε and negligible internal resistance is connected to two resistors R_1 and R_2 in series. Derive the expression for the potential difference across R_1. [3]

Answer

$I = \frac{\varepsilon}{R_1 + R_2}$ — 1

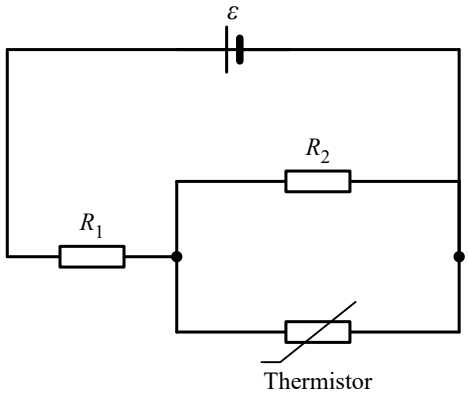
$V_1 = IR_1$ — 1

$V_1 = \frac{R_1}{R_1 + R_2}\varepsilon$ — 1

3

Explanation: The same current flows through both resistors in series.

(b) Two resistors R_1, R_2 and a thermistor are connected to a battery ε of negligible internal resistance as shown in the diagram below.



The e.m.f. of the battery is 6.0 V. The resistance of R_1 and R_2 are 2000 Ω and 5000 Ω respectively.

(i) Explain the changes in potential difference across R_1 as the temperature of the thermistor increases. [2]

Answer

As the temperature rises, the thermistor's resistance falls, so the resistance of the parallel combination falls — 1

The total resistance falls and the current rises, so the p.d. across R_1 increases — 1

2

Explanation: The p.d. across the parallel part falls correspondingly.

(ii) Determine the resistance of the thermistor when the potential difference across R_1 is 4.0 V. [2]

Answer

$I = \frac{4.0}{2000} = 2.0$ mA; p.d. across the parallel part = 2.0 V, so $R_{\text{parallel}} = 1000\ \Omega$ — 1

$\frac{1}{R_T} = \frac{1}{1000} - \frac{1}{5000} \Longrightarrow R_T = 1250\ \Omega$ — 1

2

Total for question **7**

Explanation: Kirchhoff's second law: 6.0 = 4.0 + 2.0.

Section C [30 marks]

18. (a) A copper wire has length 1.75 km and diameter 2.35 mm. A potential difference of 115 V is applied across the wire and current flows through the wire for 150.0 s.
[The resistivity of copper is 1.70×10^{-8} Ω m and electron density is 8.49×10^{28} m^{-3}.]

(i) Determine the current in the copper wire. [3]

Answer

$A = \pi(1.175 \times 10^{-3})^2 = 4.34 \times 10^{-6}$ m^2 — 1

$R = \frac{\rho L}{A} = \frac{(1.70 \times 10^{-8})(1750)}{4.34 \times 10^{-6}} = 6.86\ \Omega$ — 1

$I = \frac{115}{6.86} = 16.8$ A — 1

3

Explanation: 1.75 km = 1750 m.

(ii) Calculate the drift velocity in the copper wire. [2]

Answer

$v_d = \frac{I}{nAe} = \frac{16.8}{(8.49 \times 10^{28})(4.34 \times 10^{-6})(1.60 \times 10^{-19})}$ — 1

$v_d = 2.85 \times 10^{-4}$ m s^{-1} — 1

2

Explanation: Drift velocities are very small.

(iii) Determine the total number of electrons which passes through a point in the copper wire. [3]

Answer

$Q = It = (16.8)(150.0) = 2.51 \times 10^3$ C — 1

$N = \frac{Q}{e} = \frac{2.51 \times 10^3}{1.60 \times 10^{-19}}$ — 1

$N = 1.57 \times 10^{22}$ — 1

3

Explanation: Charge = current × time.

(b) A torch light has four dry cells which has 1.5 V each and a bulb labelled 25 W, 9.0 V. Each cell is identical, arranged in series and has internal resistance of 0.10 Ω.

(i) Determine the maximum current in the torch light. [2]

Answer

Total e.m.f. = 4 × 1.5 = 6.0 V; total internal resistance = 4 × 0.10 = 0.40 Ω — 1

Maximum current (zero external resistance) = $\frac{6.0}{0.40}$ = 15 A — 1

2

Explanation: Cells in series add their e.m.f.s and internal resistances.

(ii) Calculate the resistance of the bulb in the torch light. [2]

Answer

$R = \frac{V^2}{P} = \frac{9.0^2}{25}$ — 1

$R = 3.24\ \Omega$ — 1

2

Explanation: Use the bulb's rated values.

(iii) Calculate the current flowing through the bulb. [3]

Answer

$I = \frac{\varepsilon}{R + r}$ — 1

$I = \frac{6.0}{3.24 + 0.40}$ — 1

$I = 1.65$ A — 1

3

Total for question **15**

Explanation: The bulb runs below its rating because the supply is only 6.0 V.

19. (a) State Faraday's law and Lenz's law. [2]

Answer

Faraday: the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage — 1

Lenz: the induced current flows in a direction that opposes the change producing it — 1

2

Explanation: $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$.

(b) A 215 turns planar coil with a surface area of 25.0 cm^2 is placed in a uniform magnetic field B of 2.05 T. Initially, the plane of the coil is perpendicular to the field. The coil is then rotated about an axis perpendicular to the magnetic field with a period of 0.16 s.

(i) Determine the initial magnetic flux linkage through the coil. [3]

Answer

$N\Phi = NBA$ (plane perpendicular to the field)	1
$N\Phi = (215)(2.05)(25.0 \times 10^{-4})$	1
$N\Phi = 1.10$ Wb	1
	3

Explanation: 25.0 $cm^2 = 25.0 \times 10^{-4}$ m^2.

(ii) Derive the induced e.m.f. in the coil at time t. [4]

Answer

$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.16} = 39.3$ rad s^{-1}	1
$N\Phi = NBA \cos \omega t$	1
$\varepsilon = -\frac{d(N\Phi)}{dt} = NBA\omega \sin \omega t$	1
$\varepsilon = (1.10)(39.3) \sin 39.3t = 43.3 \sin 39.3t$ V	1
	4

Explanation: The e.m.f. is zero when the flux is maximum.

(c) A long solenoid has N_1 turns, length L and cross-sectional area A. A short coil with N_2 turns is wound around the middle of the long solenoid.

(i) Explain the phenomenon of mutual induction. [2]

Answer

A changing current in one coil produces a changing magnetic flux that links a nearby coil	1
An e.m.f. is induced in the second coil (Faraday's law), opposing the change (Lenz's law)	1
	2

Explanation: $\varepsilon_2 = -M\, dI_1/dt$.

(ii) Derive an expression for mutual inductance in terms of A, N_1, N_2 and L. [4]

Answer

Field in the solenoid: $B = \frac{\mu_0 N_1 I}{L}$ 1

The same flux $\Phi = BA$ passes through each turn of the short coil 1

Flux linkage of the coil: $N_2\Phi = \frac{\mu_0 N_1 N_2 A I}{L} = MI$ 1

$M = \frac{\mu_0 N_1 N_2 A}{L}$ 1

4

Total for question **15**

Explanation: M depends only on the geometry.

20. (a) A power source produces an alternating voltage V given by $V = 156 \sin 120\pi t$, where V is in volts and t in seconds.

(i) Determine the frequency of the alternating voltage. [3]

Answer

$\omega = 120\pi$ rad s^{-1} 1

$\omega = 2\pi f$ 1

$f = \frac{120\pi}{2\pi} = 60$ Hz 1

3

Explanation: Read ω from the equation.

(ii) Define *r.m.s. voltage*. [1]

Answer

The r.m.s. voltage is the value of the steady (d.c.) voltage that would produce the same power (heating) in a resistor 1

1

Explanation: $V_{rms} = V_0/\sqrt{2}$ for a sinusoidal voltage.

(iii) Determine the r.m.s. voltage. [2]

Answer

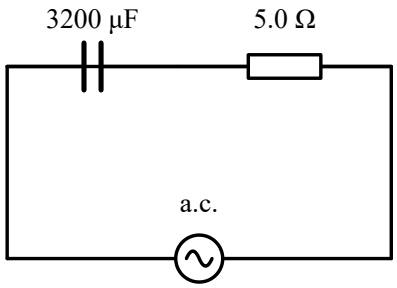
$V_{rms} = \frac{156}{\sqrt{2}}$ 1

$V_{rms} = 110$ V 1

2

Explanation: 156 V is the peak value.

(b) A power source with r.m.s. voltage 240 V and frequency 50 Hz is connected in series to a capacitor of 3200 μF and a resistor of 5.0 Ω as shown in the diagram below.



(i) Determine the impedance of the circuit. [3]

Answer

$X_C = \dfrac{1}{2\pi(50)(3200 \times 10^{-6})} = 0.995\ \Omega$ 1

$Z = \sqrt{(5.0^2 + 0.995^2)}$ 1

$Z = 5.10\ \Omega$ 1

3

Explanation: The large capacitance gives a small reactance.

(ii) Calculate the r.m.s. potential difference across the capacitor. [3]

Answer

$I_{rms} = \dfrac{240}{5.10} = 47.1\ \text{A}$ 1

$V_C = I_{rms}X_C = (47.1)(0.995)$ 1

$V_C = 46.8\ \text{V}$ 1

3

Explanation: Most of the supply voltage is across the resistor.

(iii) Determine the phase difference between the current in the circuit and the supplied voltage. [3]

Answer

$\tan \varphi = \frac{X_C}{R} = \frac{0.995}{5.0}$ 1

$\varphi = 11.3°$ 1

The current leads the supply voltage 1

3

Total for question **15**

Explanation: In an *RC* circuit the current leads.